ModernFinBERT: A Systematic Study of ModernBERT for Financial Sentiment Analysis

Yipeng Neo
neoyipeng@example.com

Abstract

We present a systematic empirical study applying ModernBERT, a recent modernized BERT architecture, to financial sentiment analysis. Through eight controlled experiments, we evaluate ModernBERT-base with LoRA fine-tuning across multiple evaluation protocols on the FinancialPhraseBank (FPB) benchmark. Our study makes several contributions: (1) we introduce a held-out evaluation protocol where FPB is entirely excluded from training, providing a stricter generalization test than standard in-domain splits; (2) we propose DataBoost, a targeted augmentation method that uses LLM-generated paraphrases of misclassified samples to improve performance; (3) we provide controlled comparisons between ModernBERT and BERT-base, finding that ModernBERT consistently matches or outperforms BERT—leading by 1.09 percentage points in 10-fold CV ($p = 0.093$) and by 7.84 points in the held-out setting on identical data—overturning an earlier spurious finding caused by a data leakage bug; and (4) we compare fine-tuned models against zero-shot Claude Opus 4.5, demonstrating that even modest fine-tuned models are orders of magnitude more cost-effective. Our results highlight the importance of evaluation protocol choice and proper experimental controls in financial NLP.

1 Introduction

Financial sentiment analysis—the task of classifying financial text as expressing positive, negative, or neutral sentiment—is a foundational capability for quantitative finance, risk management, and market surveillance [Loughran and McDonald, 2011, Malo et al., 2014]. The release of FinBERT [Araci, 2019, Yang et al., 2020] marked a significant advance, demonstrating that domain-adapted BERT models could substantially outperform general-purpose approaches on financial text.

Recently, Warner et al. [2024] introduced ModernBERT, an updated encoder architecture incorporating rotary positional embeddings (RoPE), Flash Attention, and other modern design choices. ModernBERT achieves strong performance across standard NLP benchmarks, raising the question of whether these architectural improvements translate to domain-specific tasks like financial sentiment analysis.

In this work, we conduct a systematic empirical study of ModernBERT for financial sentiment analysis. Rather than simply reporting a single accuracy number, we design eight complementary experiments that together provide a nuanced picture of model performance:

1. **Held-out evaluation** (NB01): Train on aggregated financial data *excluding* FPB, then evaluate on FPB as a truly unseen benchmark.
2. **DataBoost augmentation** (NB02): Mine validation errors and generate targeted paraphrases via Verbalized Sampling [Zhang et al., 2025] to improve difficult cases.
3. **LLM comparison** (NB03): Compare fine-tuned ModernFinBERT against Claude Opus 4.6 equipped with a purpose-built financial sentiment skill.

4. **In-domain cross-validation** (NB04): Standard 10-fold CV on FPB for comparability with prior work.
5. **Pre-trained baselines** (NB05): Compare ModernFinBERT against published FinBERT models.
6. **Multi-seed robustness** (NB06): Repeat the held-out protocol with five random seeds.
7. **Self-training** (NB07): Iterative pseudo-labeling on unlabeled financial tweets.
8. **Controlled replication** (NB09): Data contamination audit, head-to-head cross-validation, and controlled held-out comparison to isolate true architectural effects.

A key finding is that evaluation protocol choice dramatically affects reported performance: the same ModernBERT model achieves 86.88% under 10-fold CV on FPB but only 80.44% when FPB is held out entirely. Additionally, our controlled replication study reveals that an initial finding of BERT outperforming ModernBERT by 14.75 points was caused by a data leakage bug; when properly controlled, ModernBERT consistently matches or outperforms BERT. We argue that the held-out protocol provides a more realistic estimate of real-world generalization and that the financial NLP community should adopt it as a complementary evaluation standard.

2 Related Work

Financial Sentiment Analysis. The FinancialPhraseBank (FPB) dataset [Malo et al., 2014] has been the de facto benchmark for financial sentiment analysis since its release. It contains 4,846 sentences annotated by financial professionals with sentiment labels at varying agreement thresholds. Araci [2019] introduced FinBERT by further pre-training BERT on financial text, reporting 86% accuracy on FPB with 80/20 in-domain splits. Yang et al. [2020] proposed FinBERT-FinVocab with a domain-specific vocabulary, achieving 87.2% with 90/10 splits averaged over 10 runs. ProsusAI/finbert [Prosus AI, 2020] was trained directly on FPB data and is widely used as an off-the-shelf financial sentiment classifier.

A persistent challenge in this literature is the inconsistency of evaluation protocols: some works use in-domain train/test splits of varying ratios, others use cross-validation, and few evaluate on held-out data. This makes direct comparison difficult and can overstate generalization performance.

ModernBERT. ModernBERT [Warner et al., 2024] modernizes the BERT encoder architecture with several improvements: rotary positional embeddings (RoPE) replacing absolute position embeddings, Flash Attention 2 for efficient computation, GeGLU activation functions, and unpadding for variable-length sequences. With 149M parameters in the base variant, it matches or exceeds BERT-large on many benchmarks while maintaining BERT-base's inference speed.

Parameter-Efficient Fine-Tuning. Low-Rank Adaptation (LoRA; Hu et al. 2022) enables fine-tuning large models by learning low-rank updates to attention and feed-forward weights. This approach is particularly attractive when computational resources are limited, as it reduces memory requirements and training time while maintaining competitive performance.

Data Augmentation for NLP. Data augmentation has been widely used to improve NLP model robustness. Back-translation [Sennrich et al., 2016], synonym replacement [Wei and Zou, 2019], and more recently LLM-based paraphrasing [Dai et al., 2023] have shown benefits across tasks. Zhang et al. [2025] introduced Verbalized Sampling (VS), which asks LLMs to generate a *distribution* of candidate outputs with explicit probability estimates, recovering diversity that alignment-induced mode collapse normally suppresses; VS-generated synthetic data improved downstream accuracy by

up to 4.7 percentage points over standard prompting. Our DataBoost method combines VS with targeted error mining, applying diverse augmentation specifically to misclassified examples.

3 Experimental Setup

3.1 Datasets

Training Data: Aggregated Financial Sentiment Corpus. We use the `neoyipeng/financial_reasoning_a` dataset from HuggingFace, which aggregates multiple financial sentiment datasets into a unified format. After filtering for the sentiment task, the dataset contains approximately 9,603 samples split into train (8,643), validation (480), and test (480) subsets. Labels are three-class: NEGATIVE (0), NEUTRAL (1), and POSITIVE (2). Crucially, we *exclude* all FinancialPhraseBank samples (source ID 5) from training, reserving FPB as a held-out evaluation benchmark.

Evaluation Data: FinancialPhraseBank. We evaluate on two FPB subsets:

- **sentences_50agree**: 4,846 sentences where $\geq$50% of annotators agree. Label distribution: NEGATIVE 604 (12.5%), NEUTRAL 2,879 (59.4%), POSITIVE 1,363 (28.1%).
- **sentences_allAgree**: 2,264 sentences with unanimous annotator agreement. Label distribution: NEGATIVE 303 (13.4%), NEUTRAL 1,391 (61.4%), POSITIVE 570 (25.2%).

Unlabeled Data. For self-training experiments (Section 4.7), we source unlabeled financial text from the Twitter Financial News datasets on HuggingFace, filtering to 5–50 word sentences and deduplicating against all labeled data.

3.2 Model Configuration

Base Architecture. We use `answerdotai/ModernBERT-base` (149M parameters) as our primary architecture, with `bert-base-uncased` (110M parameters) as a controlled comparison.

LoRA Configuration. All fine-tuned models use LoRA with rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05, and bias set to "none." For ModernBERT, we target modules `Wqkv`, `out_proj`, `Wi`, and `Wo`. For BERT-base, we target the standard query, key, value, and output projection layers.

Training Hyperparameters. We use AdamW optimization with learning rate 2×10^{-4}, cosine learning rate schedule with 10 warmup steps, effective batch size 32 (batch size 8 with gradient accumulation factor 4), and FP16 mixed precision. Unless otherwise stated, we train for 10 epochs with early stopping based on validation loss.

3.3 Evaluation Metrics

We report accuracy and macro-averaged F1 score across all three classes. We additionally report per-class precision, recall, and F1 where relevant. For cross-validation experiments, we report mean $\pm$ standard deviation across folds.

4 Experiments and Results

4.1 Experiment 1: Held-Out Evaluation

Protocol. We fine-tune ModernBERT-base on the aggregated corpus with all FPB samples excluded, then evaluate on the full FPB dataset. This protocol tests whether the model can generalize to financial text it has never seen during training—a stricter test than the in-domain splits used in most prior work.

Results. Table 1 presents the held-out evaluation results. ModernFinBERT achieves 80.44% accuracy and 77.05% macro F1 on the challenging `sentences_50agree` split, rising to 92.98% accuracy on the easier `sentences_allAgree` split where annotators unanimously agree.

Table 1: Held-out evaluation: ModernBERT-base trained on aggregated data (FPB excluded).

Evaluation Set	Accuracy	Macro F1
FPB sentences_50agree	0.8044	0.7705
FPB sentences_allAgree	0.9298	0.9148
Aggregated test set	0.7771	0.6958

The gap between the two FPB subsets (80.44% vs. 92.98%) is informative: sentences with high annotator agreement are inherently less ambiguous, and the model handles these well. The lower performance on ambiguous sentences reflects genuine difficulty in the task rather than model failure.

4.2 Experiment 2: DataBoost — Targeted LLM Augmentation

Motivation. Rather than augmenting the entire training set uniformly, we hypothesize that targeted augmentation of *difficult* examples—those the model misclassifies—will yield more efficient improvements.

Protocol.

1. Train a baseline model on the aggregated dataset.
2. Run inference on the validation set and collect all misclassified samples.
3. For each misclassified sample, generate diverse paraphrases using Claude with Verbalized Sampling (VS-CoT; Zhang et al. 2025), preserving the *correct* (gold) label. Unlike standard paraphrasing, which mode-collapses to near-identical rewrites, VS-CoT asks the model to produce k candidate texts with explicit probability estimates, recovering distributional diversity that alignment normally compresses away. Each candidate is grounded in a different financial sub-domain (earnings calls, analyst notes, press releases, SEC filings, etc.), ensuring the augmented data spans the register diversity of real financial text.
4. Retrain on the original data augmented with these paraphrases.

Results. The baseline model misclassified 82 of 480 validation samples (17.1% error rate), yielding 246 generated paraphrases. Table 2 shows the impact of DataBoost augmentation.

DataBoost yields a 2.9 percentage point accuracy improvement and 7.8 point macro F1 improvement on the held-out test set. The disproportionate F1 gain suggests the augmentation particularly helps minority classes (NEGATIVE and POSITIVE), where misclassification is more common.

Table 2: DataBoost results on the aggregated test set (480 samples).

Model	Accuracy	Macro F1
Baseline	0.7729	0.6660
DataBoosted	0.8021	0.7443
Δ	**+0.0292**	**+0.0784**

4.3 Experiment 3: Comparison with Claude Opus 4.6

Protocol. We evaluate Claude Opus 4.6 using a purpose-built *financial sentiment engine* skill—a structured prompt encoding the three-label taxonomy, investor-lens definitions, text-length-dependent decision framework (direct classification for short text, label-first reasoning for medium text), and calibrated probability guidelines. Following Vamvourellis and Mehta [2025], the skill uses direct classification rather than chain-of-thought reasoning on short financial text, as reasoning causes "overthinking" that degrades performance on obvious sentiment signals. The evaluation is **blind**: four independent Claude Opus 4.6 agents each classify a chunk of the test set with access only to the text, not the ground truth labels.

We evaluate on a diverse 723-sample financial sentiment test set spanning FinancialPhraseBank sentences, earnings call Q&A transcripts, social media posts, press releases, and mining/resource announcements.

Results. Table 3 compares the fine-tuned ModernFinBERT against Claude Opus 4.6 with the financial sentiment skill.

Table 3: Fine-tuned ModernFinBERT vs. Claude Opus 4.6 with financial sentiment skill. ModernFinBERT evaluated on FPB sentences_50agree; Claude evaluated on a diverse 723-sample financial test set including FPB sentences.

Model	Accuracy	Macro F1	Cost/sample	Latency/sample
ModernFinBERT (149M)	0.8044	0.7705	∼$0.00001	∼2ms
Claude Opus 4.6 + skill	0.7280	0.7030	∼$0.008	∼1–2s
Fine-tuned advantage			800× cheaper	500–1000× faster

Claude Opus 4.6 with the financial sentiment skill achieves 72.8% accuracy and 0.703 macro F1—a substantial improvement over naïve zero-shot prompting but still 7.6 percentage points below the fine-tuned ModernFinBERT. Per-class analysis reveals that the POSITIVE ↔ NEUTRAL boundary is the dominant confusion source (131 of 197 errors), while NEGATIVE recall is strongest at 0.819. Despite deploying a carefully engineered prompt with domain-specific classification rules, the fine-tuned 149M-parameter model remains approximately 800× cheaper and 500–1000× faster per inference, underscoring the continued value of task-specific fine-tuning for high-volume financial applications.

4.4 Experiment 4: In-Domain Cross-Validation

Protocol. To enable comparison with prior work that uses in-domain evaluation, we perform stratified 10-fold cross-validation on FPB `sentences_50agree`. Each fold uses a 90/10 train/test

split with fresh model initialization, matching the protocol of Yang et al. [2020].

Results. Table 4 presents per-fold and summary results.

Table 4: 10-fold cross-validation on FPB sentences_50agree (4,846 samples).

Fold	Accuracy	Macro F1	$F1_{NEG}$	$F1_{NEU}$	$F1_{POS}$
1	0.8639	0.8438	—	—	—
2	0.8866	0.8765	—	—	—
3	0.8619	0.8529	—	—	—
4	0.8619	0.8494	—	—	—
5	0.8825	0.8664	—	—	—
6	0.8701	0.8718	—	—	—
7	0.8554	0.8323	—	—	—
8	0.8719	0.8441	—	—	—
9	0.8678	0.8469	—	—	—
10	0.8657	0.8562	—	—	—
Mean ± Std	**0.8688 ± 0.0096**	**0.8540 ± 0.0139**	0.8556 ± 0.0355	0.8984 ± 0.0082	0.8081 ± 0.0217

The 10-fold CV accuracy of 86.88% is comparable to published FinBERT results (86–87%) under similar evaluation protocols. The low standard deviation (±0.96%) indicates stable performance across folds. The per-class F1 breakdown reveals that NEUTRAL classification is strongest (89.84%), while POSITIVE (80.81%) and NEGATIVE (85.56%) classes are more challenging, consistent with the class imbalance.

4.5 Experiment 5: Pre-Trained FinBERT Baselines

Protocol. We evaluate two publicly available FinBERT models on FPB to establish reference baselines:

1. **ProsusAI/finbert**: Pre-trained FinBERT evaluated on FPB. This model was trained *on* FPB data, so this is effectively in-domain evaluation.
2. **yiyanghkust/finbert-tone**: FinBERT variant trained on analyst reports, evaluated zero-shot on FPB.

Results. Table 5 presents these baselines alongside our ModernFinBERT from Experiment 1. The controlled comparison between ModernBERT and BERT-base under identical conditions is deferred to Experiment 8 (Section 4.8).

Table 5: Pre-trained FinBERT baselines compared to ModernFinBERT on FPB.

Model	Protocol	FPB 50agree		FPB allAgree	
		Acc	F1	Acc	F1
ProsusAI/finbert	In-domain†	**0.8896**	**0.8825**	**0.9717**	**0.9625**
finbert-tone	Zero-shot	0.7914	0.7530	0.9152	0.8939
ModernBERT + LoRA	Held-out	0.8044	0.7705	0.9298	0.9148

† ProsusAI/finbert was trained on FPB data, so this is effectively in-domain evaluation.

ModernFinBERT's held-out accuracy (80.44%) is lower than ProsusAI/finbert's in-domain result (88.96%), as expected given that our model never sees FPB during training. On `sentences_allAgree`, however, the gap narrows substantially (92.98% vs. 97.17%), indicating that ModernFinBERT handles unambiguous financial text well even without FPB exposure.

4.6 Experiment 6: Multi-Seed Robustness

Protocol. We repeat the held-out evaluation (Experiment 1) with five random seeds (3407, 42, 123, 456, 789) to establish confidence intervals and assess training stability.

Results. Table 6 summarizes the multi-seed results.

Table 6: Multi-seed robustness analysis (5 seeds) for ModernBERT held-out evaluation.

Evaluation Set	Mean Accuracy	Mean Macro F1
FPB sentences_50agree	0.8044 ± 0.0089	0.7705 ± 0.0098
FPB sentences_allAgree	0.9298 ± 0.0025	0.9148 ± 0.0035
Aggregated test set	0.7771 ± 0.0053	0.6958 ± 0.0062

Results are stable across seeds, with standard deviations below 1% for all metrics. This confirms that ModernBERT's held-out performance is reproducible and not sensitive to random initialization.

4.7 Experiment 7: Self-Training with Pseudo-Labels

Protocol. We apply iterative self-training to leverage unlabeled financial text:

1. Train a teacher model using the DataBoosted training set from Experiment 2.
2. Source approximately 50,000 unlabeled financial sentences from Twitter Financial News datasets, filtered to 5–50 words and deduplicated against all labeled data.
3. For each self-training round, select pseudo-labeled samples using per-class top-k confidence thresholds, with increasing thresholds across rounds (15%, 25%, 40%) to progressively incorporate more data.
4. Train a fresh student model on the combined labeled and pseudo-labeled data.
5. Repeat for 3 rounds.

This per-class top-k selection strategy prevents majority class (NEUTRAL) domination of the pseudo-labeled pool, ensuring balanced representation across all three sentiment classes. Additionally, each round trains a fresh student (no weight inheritance from the teacher) and monitors validation accuracy for early stopping across rounds.

Results. After filtering and deduplication, the unlabeled pool contained approximately 30,000 financial tweets. The DataBoosted teacher labeled these with high confidence: all three classes showed mean and minimum confidence ≈1.0 among the selected top-15% samples. In Round 1, 3,217 pseudo-labeled samples were added to the training set.

Table 7 presents the full progression from baseline through DataBoost to self-training.

Self-training did not improve over the DataBoosted teacher: validation accuracy decreased from 83.54% to 82.92% in Round 1, triggering early stopping. The DataBoosted model remains the best-performing variant, with +1.65 percentage points on FPB `sentences_50agree` and +1.90 points on FPB `sentences_allAgree` over the baseline.

Table 7: Self-training progression: Baseline → DataBoosted → Self-Training Round 1.

Stage	FPB 50agree		FPB allAgree		Agg Test	
	Acc	F1	Acc	F1	Acc	F1
Baseline	0.8091	0.7810	0.9324	0.9192	0.7979	0.7258
DataBoosted	**0.8256**	**0.8052**	**0.9514**	**0.9417**	**0.8083**	**0.7633**
SelfTrain R1	0.8054	0.7700	0.9408	0.9294	0.7833	0.7187

Analysis. The failure of self-training is informative. The near-perfect confidence scores suggest the teacher's predictions on the unlabeled pool were highly polarized, producing pseudo-labels that reinforced existing decision boundaries rather than expanding them. Furthermore, the domain mismatch between the unlabeled data (informal financial tweets) and the evaluation data (formal press-release-style FPB sentences) likely diluted the training signal. This result highlights a known limitation of self-training: it is most effective when unlabeled data comes from the same domain as the target task [Xie et al., 2020].

4.8 Experiment 8: Controlled Replication Study

Motivation. An earlier version of our experiments (not reported here) appeared to show BERT-base outperforming ModernBERT by 14.75 percentage points in the held-out setting. However, we discovered a data leakage bug: the source ID filter intended to exclude FPB from training was using the wrong value (`source_id` $\neq$ `6` instead of $\neq$ `5`), causing FPB samples to leak into the training set for those runs. We conduct three sub-experiments with verified-clean data to establish the true architectural comparison.

Experiment 8a: Data Contamination Audit. Before comparing architectures, we verify that the aggregated training data contains no FPB samples that could inflate held-out results. We apply three progressively sensitive checks: (1) exact string matching between all training samples and FPB, (2) fuzzy matching using SequenceMatcher with a threshold of 0.90, and (3) semantic similarity using sentence-transformer embeddings with a cosine threshold of 0.95. All three checks return **zero matches**, confirming the training data is clean and that no data leakage contributes to the held-out evaluation results.

Experiment 8b: Head-to-Head 10-Fold Cross-Validation. We run stratified 10-fold CV on FPB `sentences_50agree` for both BERT-base and ModernBERT under identical LoRA $r = 16$ configurations (batch size 16, learning rate 2×10^{-4}, cosine schedule, 10 epochs, FP16). This eliminates any training data confound since both models train and evaluate on the same FPB folds.

ModernBERT wins 7 of 10 folds, with a mean advantage of +1.09 percentage points. A paired t-test yields $t = 1.88$, $p = 0.093$—not statistically significant at $p < 0.05$ but marginally so, suggesting a trend in ModernBERT's favor. Importantly, there is *no evidence* that BERT outperforms ModernBERT when both see identical data.

Experiment 8c: Controlled Held-Out Evaluation. We retrain both models on the *same* 8,643-sample aggregated dataset (the original version used for the ModernBERT experiments) under identical hyperparameters (batch size 8, gradient accumulation 4, learning rate 2×10^{-4}, cosine schedule, 10 epochs, seed 3407).

Table 8: Head-to-head 10-fold CV on FPB sentences_50agree. Both models use LoRA $r = 16$.

Fold	**BERT-base**	**ModernBERT**	**Δ**
0	0.8557	0.8763	+0.0206
1	0.8227	0.8474	+0.0247
2	0.8536	0.8639	+0.0103
3	0.8619	0.8330	−0.0289
4	0.8701	0.8680	−0.0021
5	0.8515	0.8619	+0.0103
6	0.8678	0.8636	−0.0041
7	0.8657	0.8843	+0.0186
8	0.8574	0.8884	+0.0310
9	0.8202	0.8492	+0.0289
Mean ± Std	**0.8527 ± 0.0175**	**0.8636 ± 0.0171**	**+0.0109**

Table 9: Controlled architecture comparison on FPB (held-out, FPB excluded from training). Both LoRA models trained on identical 8,643 samples with identical hyperparameters. Reference baselines from Experiment 5 included for context.

Model	**Protocol**	**FPB 50agree**		**FPB allAgree**	
		Acc	Macro F1	Acc	Macro F1
Reference baselines (Experiment 5):					
ProsusAI/finbert	In-domain†	0.8896	0.8825	0.9717	0.9625
finbert-tone	Zero-shot	0.7914	0.7530	0.9152	0.8939
Controlled comparison (identical 8,643-sample training set):					
ModernBERT + LoRA	Held-out	**0.8093**	**0.7793**	**0.9329**	—
BERT-base + LoRA	Held-out	0.7309	0.6051	0.8366	—
Δ (MB − BERT)		**+0.0784**	**+0.1742**	**+0.0963**	

† Trained on FPB data (in-domain evaluation).

On identical training data, **ModernBERT outperforms BERT-base by 7.84 percentage points** on FPB `sentences_50agree` and 9.63 points on `sentences_allAgree`. ModernBERT also approaches ProsusAI/finbert (80.93% vs. 88.96%) despite never seeing FPB during training, while BERT-base falls well short (73.09%).

5 Analysis and Discussion

5.1 The Protocol Gap

Our experiments reveal a fundamental tension in financial sentiment evaluation. The same ModernBERT model achieves vastly different performance depending on the evaluation protocol:

- **10-fold CV on FPB**: 86.88% accuracy (comparable to published FinBERT results)
- **Held-out evaluation**: 80.44% accuracy (FPB excluded from training)

This 6.4 percentage point "protocol gap" arises because in-domain CV allows the model to

train on FPB-distributed data, while the held-out setting tests true generalization. We argue that reporting both protocols is essential for honest evaluation, and that the held-out number better reflects real-world deployment performance where the model encounters novel financial text.

5.2 Architecture Comparison: ModernBERT vs. BERT

Our controlled replication (Experiment 8) overturns the initial finding that BERT outperforms ModernBERT. The apparent 14.75-point gap was caused by a source ID filtering bug that allowed FPB to leak into the BERT training set. When properly controlled on verified-clean data:

- **Held-out evaluation**: ModernBERT outperforms BERT by 7.84 points on identical 8,643-sample training data (Table 9).
- **10-fold CV**: ModernBERT leads by 1.09 points, $p = 0.093$ (Table 8).

Across both evaluation protocols, ModernBERT is consistently competitive with or superior to BERT-base. The held-out advantage (+7.84 points) suggests that ModernBERT's modernized architecture—rotary positional embeddings, Flash Attention, and GeGLU activations—provides genuinely stronger representations for financial text, even without domain-specific pre-training. This is encouraging for practitioners, as it suggests that upgrading from BERT to ModernBERT yields improvements on domain-specific tasks without requiring additional pre-training.

This finding also serves as a cautionary tale about the importance of data pipeline verification. The original confound—a single wrong constant in a filter—produced a compelling but entirely spurious result that would have led to incorrect conclusions about architecture design.

5.3 Fine-Tuning vs. Skill-Prompted LLMs

The comparison with Claude Opus 4.6 (Section 4.3) demonstrates that task-specific fine-tuning remains strongly preferable for production financial sentiment analysis, even when the LLM is equipped with a purpose-built financial sentiment skill:

- **Accuracy**: ModernFinBERT (80.44%) outperforms Claude Opus 4.6 with a carefully engineered financial sentiment skill (72.8%) by 7.6 percentage points.
- **Cost**: Fine-tuned inference costs ∼800× less per sample.
- **Latency**: Fine-tuned models are 500–1000× faster, enabling real-time processing.
- **Consistency**: Fine-tuned models produce deterministic outputs; LLM outputs vary across runs.

Notably, the Claude evaluation used a structured skill with domain-specific classification rules, investor-lens label definitions, and direct classification (avoiding chain-of-thought overthinking per Vamvourellis and Mehta 2025)—not a naïve zero-shot prompt. This represents a strong-effort LLM baseline. The persistent accuracy gap suggests that fine-tuning provides pattern recognition at decision boundaries that prompt engineering alone cannot replicate. LLMs remain valuable for financial analysis tasks requiring reasoning, explanation, or handling novel formats, but for high-volume three-class sentiment classification, fine-tuning remains the pragmatic choice.

5.4 DataBoost: Targeted Augmentation with Verbalized Sampling

The DataBoost method's 7.8-point macro F1 improvement (Table 2) from only 246 generated samples is notable. Compared to uniform augmentation strategies that generate paraphrases for all training samples, DataBoost focuses computational resources where they matter most—on the decision boundaries where the model struggles.

The use of Verbalized Sampling (VS-CoT) for paraphrase generation is a key design choice. Standard LLM paraphrasing mode-collapses to near-identical rewrites of the input [Zhang et al.,

2025], which provides limited additional training signal. By requiring the generator to produce k candidates with explicit probability estimates spanning different financial registers, VS-CoT ensures that each misclassified sample yields genuinely diverse augmentations—from earnings call transcripts to analyst notes to SEC filings—rather than superficial lexical variation.

The disproportionate F1 improvement (7.8 points) relative to accuracy improvement (2.9 points) confirms that DataBoost primarily helps minority classes. This is valuable in financial applications where detecting negative sentiment (e.g., risk signals) is often more important than correctly classifying the dominant neutral class.

5.5 Self-Training: When More Data Hurts

The negative self-training result (Section 4.7) provides a useful cautionary finding. Despite careful anti-confirmation-bias measures—per-class selection, fresh student initialization, and early stopping—adding 3,217 pseudo-labeled tweets degraded performance across all metrics.

Two factors likely contribute. First, the **domain mismatch**: Twitter financial text is informal, abbreviated, and opinionated, while FPB consists of formal press-release sentences. Pseudo-labels from tweets may teach the model stylistic patterns that are counterproductive for formal text classification. Second, the **overconfident teacher**: with mean and minimum confidence both near 1.0, the selected pseudo-labels carried no meaningful uncertainty signal, effectively amplifying the teacher's existing biases rather than adding complementary information.

This suggests that for self-training to succeed in financial NLP, practitioners should prioritize domain-matched unlabeled data and consider calibration techniques to ensure meaningful confidence differentiation.

6 Limitations

- **Single benchmark**: While FPB is the standard financial sentiment benchmark, evaluating on additional datasets (SemEval Financial, analyst reports) would strengthen our conclusions.
- **Base models only**: We evaluate only base-sized models (110–149M parameters). Results may differ with large variants.
- **English only**: All experiments use English financial text. Financial NLP in other languages remains unexplored.
- **LoRA only**: Full fine-tuning could yield different relative performance between architectures, particularly since ModernBERT's fused QKV projections receive effectively lower per-component rank than BERT's separate Q, K, V projections under the same LoRA rank.
- **Training data composition**: The aggregated training corpus combines heterogeneous sources. The interaction between source diversity and model architecture deserves further study.
- **Single training data size**: The controlled BERT vs. ModernBERT comparison uses only 8,643 training samples. A sample efficiency study across multiple dataset sizes would reveal how the architectural advantage changes with data scale.
- **Self-training data**: The unlabeled pool consisted solely of financial tweets, which differ stylistically from FPB's press-release sentences. Self-training with domain-matched unlabeled data (e.g., financial news headlines) may yield different results.

7 Conclusion

We have presented a systematic study of ModernBERT for financial sentiment analysis, encompassing eight controlled experiments. Our key findings are:

1. **Evaluation protocol matters**: The 6.4-point gap between in-domain CV and held-out evaluation highlights the need for standardized, rigorous evaluation in financial NLP.
2. **Data pipeline verification is essential**: An initial finding that BERT outperforms ModernBERT by 14.75 points was entirely caused by a data leakage bug (wrong source ID filter). Controlled experiments on verified-clean data show ModernBERT consistently matches or outperforms BERT.
3. **DataBoost works**: Targeted LLM augmentation of misclassified examples yields efficient improvements, particularly for minority classes.
4. **Fine-tuning beats skill-prompted LLMs**: Even modest fine-tuned models outperform state-of-the-art LLMs equipped with purpose-built classification skills on structured sentiment tasks, while being orders of magnitude cheaper and faster.
5. **Self-training requires domain match**: Iterative pseudo-labeling on out-of-domain tweets did not improve performance, highlighting the importance of domain-matched unlabeled data for semi-supervised methods.

These findings provide practical guidance for practitioners: ModernBERT is a strong default for financial sentiment analysis, use held-out evaluation to honestly assess generalization, verify data pipelines carefully before drawing architectural conclusions, and consider targeted augmentation for difficult cases.

The ModernFinBERT model is available at `neoyipeng/ModernFinBERT-base` on HuggingFace. All experiment code and notebooks are publicly released to support reproducibility.

References

Dogu Araci. Finbert: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*, 2019.

Haixing Dai, Zhengliang Liu, Wenxiong Liao, Xiaoke Huang, Yihan Cao, Zhi Wu, Lin Zhao, Shaochen Xu, Wei Liu, Ningkui Liu, et al. Chataug: Leveraging chatgpt for text data augmentation. *arXiv preprint arXiv:2302.13007*, 2023.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shanen Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2022.

Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of Finance*, 66(1):35–65, 2011.

Pekka Malo, Ankur Sinha, Pekka Korhonen, Jyrki Wallenius, and Pyry Takala. Good debt or bad debt: Detecting semantic orientations in economic texts. In *Journal of the Association for Information Science and Technology*, volume 65, pages 782–796, 2014.

Prosus AI. Prosusai/finbert. https://huggingface.co/ProsusAI/finbert, 2020. HuggingFace model repository.

Rico Sennrich, Barry Haddow, and Alexandra Birch. Improving neural machine translation models with monolingual data. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pages 86–96, 2016.

Konstantinos Vamvourellis and Vikas Mehta. When chain-of-thought hurts: Direct classification outperforms reasoning on short financial text. *arXiv preprint*, 2025.

Benjamin Warner, Antoine Hundt, Quentin Lhoest, Clemens Schreiber, Oskar Van Wyk, Aakanksha Chowdhery, and Jonathan Kew. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

Jason Wei and Kai Zou. Eda: Easy data augmentation techniques for boosting performance on text classification tasks. *arXiv preprint arXiv:1901.11196*, 2019.

Qizhe Xie, Minh-Thang Luong, Eduard Hovy, and Quoc V Le. Self-training with noisy student improves imagenet classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10687–10698, 2020.

Yi Yang, Mark Christopher Siy Uy, and Allen Huang. Finbert: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*, 2020.

Ziqi Zhang, Jingyu Zhao, et al. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity. *arXiv preprint arXiv:2510.01171*, 2025.